

Kai-Jie (KJ) Lin

✉ linkai0508@gmail.com KJLdefeated in Kai-Jie-Lin

Education

Carnegie Mellon University

Master of Electrical and Computer Engineering

Pittsburgh, USA

Jan. 2027 – Jun. 2028

National Yang Ming Chiao Tung University

B.S. in Computer Science; GPA: 3.83/4.0

Hsinchu, Taiwan

Sept. 2021 – Jun. 2025

Technical Skills

Programming Languages: Python, C/C++, SQL, CUDA, Triton

Expertise: LLM, Multimodal LLM, Agentic-RL, Reasoning, RLHF, DPO, GRPO, RAG, Diffusion Models

Frameworks: PyTorch, TensorFlow, vLLM, SGLang, Unsloth, Apache Spark

Work & Research Experience

Maintainer @ RL-Kernel: Bitwise-Consistent Kernels for RL Post-Training

[Github, 200+ ★]

CUDA Kernel, Triton, Distributed Training Infra, RL Infra

Jun. 2026 – Now

- Code owner for the kernel tree and CUDA/Triton backends of an open-source RL post-training library targeting bitwise operator-level train-inference consistency across heterogeneous engines.
- Built a fused linear logprob operator(#122) that cuts forward memory from 5GB to near 0 and runs 2.0× faster than Triton on H100, by fusing the LM head with split-V online softmax and utilizing SM90 TMA + tensor-core.
- Built a batch-invariant log-prob kernel(#204) that is 15.4× faster than PyTorch and 1.74× faster than Triton on H100 at zero extra memory (vs. 7.1 GB), using a streaming log-sum-exp whose reduction order is independent of batch shape.

Research Assistant (Part-Time) @ NVIDIA – NYCU Advanced Research Collaboration

Hsinchu, Taiwan

LLM, RL, Post-Training, Agentic System

Jul. 2023 – Jul. 2026

- Built RL post-training infrastructure with disaggregated rollout (vLLM) and training (DeepSpeed ZeRO), implementing rollout training weight resharding and colocation, cutting GRPO wall-clock 6× (72h→12h).
- Implemented training infrastructure for multimodal post-training on Qwen2.5-VL, including LoRA fine-tuning, reward verification, and rollout/training weight synchronization.
- Developed an embodied agent system on MineDojo, integrating visual observation processing through Qwen2.5-VL and multi-turn action planning.

Machine Learning Engineer @ Rayleigh Vision Intelligence (RVI)

Hsinchu, Taiwan

Multimodal Learning, ML Pipeline, Production Deployment

Jun. 2025 – Jun. 2026

- Built and deployed an end-to-end production pipeline (Spark-based ingestion, multimodal training and batch inference) serving inline wafer quality analysis from 10 to 50 wafers/hr, replacing manual visual inspection on the production line.
- Engineered a wafer-labeling and synthetic-data pipeline that expanded the rare-defect training set 250×, cutting manual annotation effort and correcting severe class imbalance.
- Designed a multimodal inspection model fusing wafer imagery (ViT) with tabular LED signals (FT-Transformer), cutting prediction error 50% (MAE) over the prior single-modality baseline.

Publications

AskR: Learning to Ask for Intrinsic Reward Elicitation for Test-Time Reasoning

Kai-Jie Lin*, and Kuang-Da Wang*, et al.

ICLR'27 (Aiming submission)

- A test-time reasoning framework that decomposes reasoning into sub-question/sub-answer pairs and converts QA self-verification into dense intrinsic rewards.
- Trains an asker module via GRPO to generate sub-questions to induce intrinsic rewards on a frozen backbone, without modifying base model weights.
- Outperforms PRM-guided reasoning by up to 20% and RLVR-trained GRPO by 10% with far less training data on mathematical and scientific reasoning tasks.

BOFormer: A Generalized Deep Q-Learning Framework for Multi-Objective Bayesian Optimization

Kai-Jie Lin*, and Yu-Heng Hung*, et al.

ICLR'25

- A Transformer-based Q-learning framework for sample efficient Black-Box optimization, learning history-aware acquisition policies that adapt over the full trial sequence.
- Trains the Q-Transformer with prioritized trajectory replay and distributed environment sampling for stable off-policy learning across diverse BO tasks.
- Outperforms SOTA Bayesian optimization baselines by up to 20% on tuning hyperparameters of 3D Gaussian Splatting, the SOTA method for 3D scene reconstruction.